

LAB 6

Applying Meta Tags and Multimedia Tags in HTML

Subject: Web Designing

Lab Focus: HTML Meta Tags & Multimedia Embedding

Tools Required: Any text editor (VS Code / Notepad) and a Web Browser

Prerequisite Knowledge: Basic HTML structure (html, head, body tags)

Topics Covered in This Lab

1. Basic Meta Tags — description, keywords, author, viewport
2. Auto-refresh Meta Tag and Charset Meta Tag
3. Embedding Audio using the <audio> tag
4. Embedding Video using the <video> tag
5. Embedding Multimedia/External Content using <embed> and <object> tags

Lab 6: Meta Tags and Multimedia Tags

Objective: To understand the purpose of HTML meta tags in providing metadata about a webpage, and to learn how to embed multimedia content such as audio and video directly into a webpage.

What are Meta Tags?

The `<meta>` tag provides **metadata** — that is, "data about data." Meta tags do not display any visible content on the page itself; instead they give the browser, and more importantly **search engines**, extra information about the document such as its description, character encoding, author, and how the viewport should behave on different devices. All meta tags are placed inside the `<head>` section of an HTML document and are written as **empty (self-closing) elements** — they never have a closing tag.

Exercise 1: Basic Meta Tags (description, keywords, author, viewport)

Explanation

Every well-built webpage begins with a few standard meta tags that describe the page to browsers and search engines:

Meta Tag	Purpose	Example
<code>description</code>	A short summary of the page content. Search engines like Google often display this text as the snippet below the page title in search results, so it directly affects click-through rate.	<code><meta name="description" content="..."></code>
<code>keywords</code>	A comma-separated list of words relevant to the page content. Historically used for SEO; modern search engines give it little weight, but it is still taught as part of proper HTML metadata practice.	<code><meta name="keywords" content="HTML, CSS, ..."></code>
<code>author</code>	Specifies the name of the document's author/creator.	<code><meta name="author" content="..."></code>
<code>viewport</code>	Controls how the page scales and is sized on different devices (mobiles, tablets, desktops). Essential for responsive web design .	<code><meta name="viewport" content="width=device-width, initial-scale=1.0"></code>

Why the viewport tag matters: Without it, mobile browsers render the page at a desktop width (usually 980px) and then shrink it to fit the screen, making text unreadably small. The `width=device-width` value tells the browser to match the page width to the device's screen width, and `initial-scale=1.0` sets the initial zoom level.

Code Example

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <meta name="description" content="Learn HTML meta tags and multimedia embedding with practical examples">
  <meta name="keywords" content="HTML, meta tags, web designing, multimedia">
  <meta name="author" content="Web Designing Lab">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Basic Meta Tags Demo</title>
</head>
<body>
  <h1>Welcome to Web Designing Lab</h1>
  <p>This page demonstrates the use of basic meta tags.</p>
</body>
</html>
```

Output/Effect: The visible page shows only the heading and paragraph text. The meta tags themselves are invisible in the browser window, but can be seen in the page's HTML source (right-click → View Page Source), and the description text would appear in a search engine result snippet.

Exercise 2: Auto-Refresh Meta Tag and Charset Meta Tag

Explanation

a) Auto-Refresh (`http-equiv="refresh"`): The `http-equiv` attribute lets a meta tag act like an HTTP header. When set to `"refresh"`, the `content` attribute defines the number of seconds before the browser automatically reloads the page (or redirects to another URL if a `url=` value is included).

b) Character Encoding (`charset`): The `charset` meta tag declares which character set the browser should use to correctly interpret the bytes of the HTML file into readable text. `UTF-8` is the universal standard because it supports virtually every language and symbol (including emojis), so it should always be the very first tag inside `<head>`.

Code Example

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <!-- Page will auto-refresh every 5 seconds -->
  <meta http-equiv="refresh" content="5">
  <title>Auto Refresh Demo</title>
</head>
<body>
  <h1>This page refreshes automatically every 5 seconds!</h1>
  <p>Character encoding used: UTF-8</p>
</body>
</html>
```

Redirecting to another page: Writing `content="5; url=https://example.com"` refreshes to a new URL after 5 seconds instead of reloading the same page — commonly used for "This page has moved" notices.

Output/Effect: The browser tab will visibly reload (a small loading spinner appears) every 5 seconds, repeating for as long as the page stays open.

Exercise 3: Embedding Audio using the `<audio>` Tag

Explanation

The `<audio>` tag (introduced in HTML5) embeds sound content — music, podcasts, sound effects — without needing external plug-ins like Flash. It supports the following key attributes:

Attribute Description	
<code>controls</code>	Displays the browser's built-in play/pause, volume, and seek-bar UI.
<code>autoplay</code>	Starts playing the audio automatically as soon as the page loads (most modern browsers block autoplay with sound unless the tag is also <code>muted</code>).
<code>loop</code>	Automatically restarts the audio from the beginning once it finishes playing.
<code>src</code>	Specifies the path/URL of the audio file directly on the tag (alternative to using a nested <code><source></code> tag).
<code><source></code>	A child element used to provide multiple audio formats (e.g. <code>.mp3</code> , <code>.ogg</code>) so the browser picks the one it supports.

Code Example

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Audio Tag Demo</title>
</head>
<body>
  <h2>Playing Background Music</h2>
  <audio controls autoplay loop>
    <source src="music.mp3" type="audio/mpeg">
    <source src="music.ogg" type="audio/ogg">
    Your browser does not support the audio element.
  </audio>
</body>
</html>
```

Important: The text between the opening and closing `<audio>` tags ("Your browser does not support...") is **fallback content** — it only displays if the browser cannot play HTML5 audio at all.

Output/Effect: A horizontal audio player control bar appears on the page with play/pause, a progress/seek bar, volume control, and a download option. Playback begins immediately and loops continuously.

Exercise 4: Embedding Video using the `<video>` Tag

Explanation

The `<video>` tag embeds video content natively in the browser. Like `<audio>`, it is a paired tag and supports several useful attributes:

Attribute	Description
<code>controls</code>	Shows the standard video player controls (play, pause, seek, volume, fullscreen).
<code>poster</code>	Specifies an image to display as the video's placeholder/thumbnail before the user presses play.
<code>width / height</code>	Sets the display dimensions of the video player in pixels.
<code><source></code> (multiple)	Lets you provide the same video in several formats (.mp4, .webm, .ogg) so different browsers can each pick a format they can decode.
<code>autoplay / loop / muted</code>	Same behaviour as in the audio tag — automatically start playing, replay after finishing, and play with sound muted.

Code Example

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Video Tag Demo</title>
</head>
<body>
  <h2>Sample Video Player</h2>
  <video width="480" height="270"
    controls poster="thumbnail.jpg">
    <source src="movie.mp4" type="video/mp4">
    <source src="movie.webm" type="video/webm">
    Your browser does not support the video tag.
  </video>
</body>
</html>
```

Output/Effect: A 480×270 pixel video frame is displayed showing the `thumbnail.jpg` poster image with a play button overlay. Clicking play loads and plays `movie.mp4` (or the next supported format) with full player controls beneath it.

Exercise 5: Embedding External Content — `<embed>` and `<object>` Tags

Explanation

Besides `<audio>` and `<video>`, HTML provides two more general-purpose tags for embedding external resources — such as PDFs, Flash movies (legacy), or other multimedia — directly inside a webpage:

Tag	Description	Key Attributes
<code><embed></code>	A simple, self-closing (void) element used to embed external content/plugin based content such as PDFs, audio, or video. It has minimal built-in fallback support.	<code>src</code> , <code>type</code> , <code>width</code> , <code>height</code>
<code><object></code>	A more powerful, paired container tag that can embed images, PDFs, videos, or even other HTML pages. It supports fallback content between its opening and closing tags, which displays if the browser cannot render the resource.	<code>data</code> , <code>type</code> , <code>width</code> , <code>height</code>

`<embed>` vs `<object>`: The key difference is that `<embed>` is self-closing with no fallback content, while `<object>` is a container tag that can hold fallback text/HTML for browsers that cannot display the embedded resource — making `<object>` the more robust and standards-preferred choice.

Code Example

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Embed and Object Tag Demo</title>
</head>
<body>
  <h2>Using the <embed> tag</h2>
  <embed src="sample.pdf" type="application/pdf"
    width="500" height="300">

  <h2>Using the <object> tag</h2>
  <object data="sample.pdf" type="application/pdf">
```

```

        width="500" height="300">
        <p>Your browser cannot display this PDF. <a href="sample.pdf">Download it here</a></p>
    </object>
</body>
</html>

```

Output/Effect: Both sections display an embedded PDF viewer, 500×300 pixels in size, showing the first page of "sample.pdf" with the browser's native PDF controls. If the <object> version cannot render the PDF, it instead shows the fallback paragraph with a download link.

Quick Revision — Lab 6 Summary

Tag / Attribute	One-line Purpose
<meta name="description">	Short page summary for search engines
<meta name="keywords">	List of relevant SEO keywords
<meta name="author">	Names the page's creator
<meta name="viewport">	Controls responsive scaling on devices
<meta http-equiv="refresh">	Auto-reloads/redirects the page after N seconds
<meta charset="UTF-8">	Declares character encoding of the document
<audio>	Embeds playable sound with controls/autoplay/loop
<video>	Embeds playable video with controls/poster/multiple sources
<embed>	Simple void tag to embed external plug-in content
<object>	Container tag to embed content with fallback support

Viva / Practice Questions

1. What is the difference between the `keywords` and `description` meta tags?
2. Why must the viewport meta tag be used for responsive websites?
3. What does `http-equiv="refresh"` do, and how do you redirect to a different URL with it?
4. Why should the charset meta tag usually be the first line inside <head>?
5. List three attributes of the <audio> tag and explain each.
6. What is the purpose of the `poster` attribute in the <video> tag?
7. Why is more than one <source> tag often provided inside <audio>/<video>?
8. Differentiate between the <embed> and <object> tags.